Session Objectives

- Describe and discuss the pathogenesis of **asbestos** related lung disease as it relates to diffuse interstitial fibrosis and correlate the clinical and pathological (gross and microscopic) features
- Briefly describe the development of benign and malignant pleural lesions in relation to asbestos exposure
- Describe the clinical and pathological features of **berylliosis** and discuss its immune pathogenesis
- Discuss, describe, and explain the pathogenesis and various forms of **coal workers pneumoconiosis** as well as the clinical, microscopic, and gross features
- Discuss, describe, and explain the pathogenesis and histological profile of **silicosis** with clinical correlation and delineation of various manifestations

Asbestos - Background

- Asbestosis - pneumoconiosis caused by inhalation of asbestos fibers.
- Characterized by slowly progressive, diffuse pulmonary fibrosis
- Pulmonary disorders associated with asbestos exposure include:
  - Asbestosis (Interstitial fibrosis)
  - Pleural disease (focal and diffuse benign pleural plaques), pleural effusions
  - Malignancies (non-small cell and small cell carcinoma of the lung as well as malignant mesothelioma)

Robbins and Cotran, Pathological Basis of Disease, 11th ed., 2026, Ch. 15

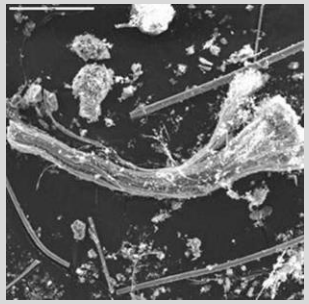
Asbestos - Background

Naturally occurring fibers - hydrated magnesium silicates

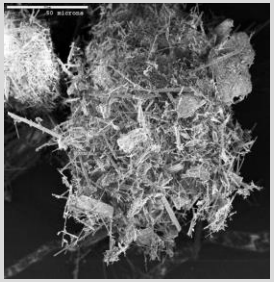
- Tensile strength, chemical properties suited for construction and insulation.
- **Asbestos fibers - two categories based on shape**
  - **Serpentine fibers** - long, curly strands (eg. chrysotile) - accounts for over 90 percent of the asbestos in US

    -**less toxic** than the amphibole fibers

    -Because of structure (curve), serpentine fibers more likely to get impacted in upper respiratory tract and **removed by mucociliary elevator**



  - **Amphibole fibers** - long, straight, rod-like structures. (eg. crocidolite, amosite, tremolite, and others)
    - **more pathogenic** especially for induction of mesothelioma, because of aerodynamic properties and solubility



Robbins and Cotran, Pathological Basis of Disease, 10th ed., 2020, Ch. 15

Asbestos - Clinical

- Bibasilar, fine end-inspiratory crackles
- Clubbing
- Cor pulmonale
- Peripheral edema
- Jugular venous distension
- Hepatojugular reflux
- Right ventricular heave or gallop
- May lead to respiratory failure or death







Cough, sputum production, and wheezing are unusual – if present, tend to be because of cigarette smoking rather than asbestos-induced lung disease

More advanced disease can be associated with hypoxemia at rest

Asbestos – Imaging

Chest Xray:

- Small bilateral parenchymal opacities with multinodular or reticular pattern, often with associated pleural abnormalities
- Interstitial process typically begins in lower lung zones
- Bilateral mid-lung zone plaques on parietal pleura
- Honeycombing and upper lobe involvement develop in advanced stages of disease
- Hilar and mediastinal lymphadenopathy are **not seen with asbestosis** and should suggest presence of another process.

Robbins and Cotran, Pathological Basis of Disease, 11th ed., 2026, Ch. 15

Case Question

A 75-year-old man who had worked in a shipyard dies of chronic lung disease. He had also been a smoker for 40 years (1 ppd). Autopsy reveals extensive pulmonary fibrosis, and iron stains of lung tissue show numerous ferruginous bodies. The dome of the diaphragm shows the presence of pleural plaques. What is the most appropriate diagnosis?

A. Anthracosis

B. Sarcoidosis

C. Tuberculosis

D. Asbestosis

E. Silicosis

Berylliosis - Pathogenesis

- **Acute beryllium disease** in humans:
  - attributed to toxic, dose-related lung injury
  - characterized by acute inflammatory reactions in upper airways
  - bronchiolitis, pulmonary edema, and chemical pneumonitis
- **Chronic beryllium disease - multisystem** disorder characterized by noncaseating (non-necrotizing) **granulomas** that occur throughout the body; primary manifestation is in the lung, esp. affects upper lobes
- Occurs in industries such as metal machining, ceramics, and nuclear weapons manufacturing

Berylliosis – Chronic Beryllium Disease

On histopath exam, chronic beryllium disease: characterized by lymphocytic (helper/inducer T cell) alveolitis, as well as noncaseating epithelioid granulomas indistinguishable from those of sarcoidosis

Imaging: Usual finding is ill-defined nodular or irregular opacities; hilar adenopathy seen in approximately 40% of cases, but usually mild



In **later** stages of the disease, patchy fibrosis occurs with adjacent hyperinflation or distortion and extensive **honeycombing**

Clinical features of chronic beryllium disease similar to those of pulmonary sarcoidosis:

- Dyspnea, cough, chest pain, weight loss, fatigue, and arthralgias
- Physical signs may include crackles, but signs of lung disease are often absent

Case Question

A 69-year-old man who works in a steel yard reports to the office with a chronic cough and weight loss. On CT, irregular opacities are found, along with mild hilar adenopathy. The pulmonologist decides to biopsy the lung which reveals noncaseating epithelioid granulomas. What is the most appropriate diagnosis?

A. Silicosis

B. Chronic Obstructive Pulmonary Disease

C. Berylliosis

D. Asbestosis

E. Coal Worker's Pneumoconiosis

Coal Workers Pneumoconiosis – Progressive Massive Fibrosis

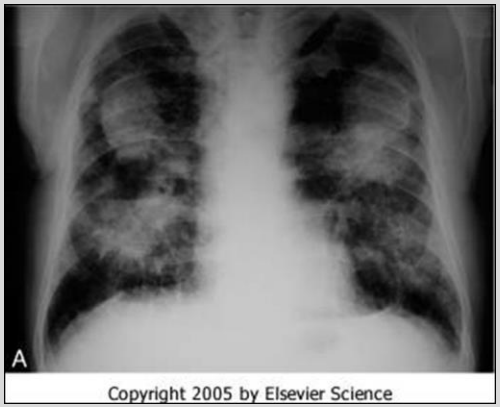
- On background of simple CWP after many years
- Multiple, intense blackened scars **greater than 1 cm (up to 10 cm)**
- Micro shows dense collagen and pigment; center of lesion may be **necrotic because of ischemia**
- PMF develops in about 10% of cases of simple CWP
- Leads to worsening lung function, pulmonary hypertension, and cor pulmonale

Source: Robbins and Cotran, Pathological Basis of Disease, 11th ed., 2026, Ch. 15

Coal Workers Pneumoconiosis – Progressive Massive Fibrosis

- Symptoms of cough and sputum reported by most coal miners are likely a result of dust-induced chronic bronchitis
- Breathlessness during exertion usually caused by associated chronic airflow limitation or by development of fibrosis
- Hallmark of simple CWP on the posteroanterior chest radiograph is presence of small rounded opacities in the lung parenchyma





Source: Robbins and Cotran, Pathological Basis of Disease, 11th ed., 2026, Ch. 15

Coal Workers Pneumoconiosis – Clinical

LUNG FUNCTION

Simple CWP does not really affect lung function

Coal dust has been shown to cause both **obstructive and restrictive** lung disease

Lung function deficits in complicated CWP (PMF) include:

- reduction in FVC and FEV_1
- increased total lung capacity and residual volume

MANAGEMENT AND CONTROL

Once established, the fibrotic process of CWP is thought to be irreversible. 6

Source: Robbins and Cotran, Pathological Basis of Disease, 11th ed., 2026, Ch. 15

Case Question

A 57-year-old man who lives and works in West Virginia reports to the office for a chronic wet cough. On chest X-ray, small rounded opacities in the lung parenchyma are noted. Based on his most likely diagnosis, where in the lung is this disease process most likely to first develop?

A. Right middle lobe
B. Right lower lobe
C. Left lower lobe
D. Left upper lobe
E. Bilaterally, in the lung base

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Silicosis

Chronic silicosis, typically follows exposure, measured in decades (10 – 30 years), to respirable dust usually containing less than 30% quartz – Most common

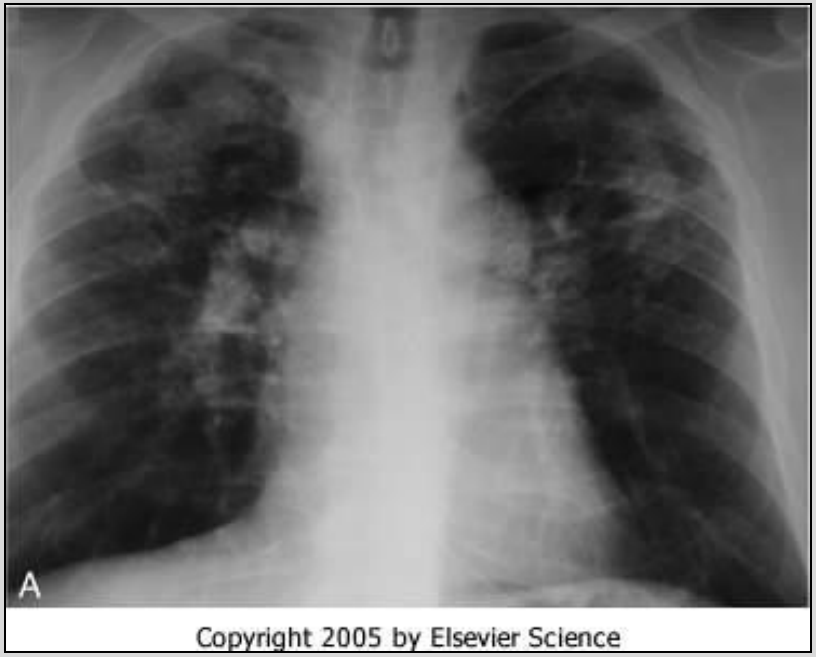
Accelerated silicosis, follows shorter, heavier exposure (within 10 years)

Acute silicosis (silicoproteinosis), follows intense exposure to fine dust of high silica content, such as that found in **sandblasting industries**, for periods measured in weeks to months rather than years – Rare

Complicated silicosis with disease progression in both chronic and accelerated silicosis, the nodules become confluent, leading to the development of **progressive massive fibrosis (PMF)**

Source: Robbins and Cotran, Pathological Basis of Disease, 11th ed., 2026, Ch. 15
Waxner, Saraceno, Goblin

Patient with chronic silicosis *may present without symptoms* for assessment of an abnormal chest radiograph



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Posteroanterior radiograph showing features of progressive massive fibrosis (PMF) **with "eggshell" calcification of the hilar nodes**, in keeping with complicated silicosis, in a man who had shoveled sand in a glass factory

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Silicosis - Clinical

LUNG FUNCTION

- PFTs may be normal but may become abnormal if progressive massive fibrosis develops
- Spirometric tests (FEV_1, FEV_1/FVC, and maximal midexpiratory flow) usually reflect airflow limitation
- Reduction in diffusing capacity is generally apparent in more advanced chronic silicosis

MANAGEMENT AND CONTROL

Once established, the fibrotic process of chronic silicosis is thought to be irreversible

Case question

A 60-year-old mason complains of shortness of breath, which has become progressively worse during the past year. A chest Xray shows small nodular shadows in both lungs. PFTs reveal a pattern consistent with restrictive lung disease. The patient subsequently develops congestive heart failure and expires. Autopsy discloses numerous small fibrotic nodules in both lungs. Histological examination of these nodules show concentric, whorling, interlacing dense, hyalinized collagen fibers with no granulomas. What is the most likely diagnosis?

A. Tuberculosis

B. Asbestosis

C. Anthracosis

D. Silicosis

E. Sarcoidosis

Summary Slide

- Definition of Pneumoconiosis – restrictive lung disease/occupations
- Asbestos-related disease – pathogenesis, clinical, lab, Xray
  - Asbestos fibers
  - Pleural plaques and effusions
  - Diffuse interstitial lung fibrosis
  - Malignant mesothelioma
- Berylliosis – pathological and immune pathogenesis
  - 17% of acute beryllium disease progress to chronic
  - Non-caseating granulomas, interstitial lung disease
  - Clinical similar to sarcoidosis
- Coal Workers' Pneumoconiosis
  - Coal macule and coal nodule (coal macule – macro/micro: aggregate of dust laden macrophages around resp. bronchiole with reticulin fibers and collagen)
  - Complicated CWP – progressive massive fibrosis – on background of simple CWP after many years
- Silicosis – chronic, accelerated, acute
  - Silicotic nodule
  - Pathogenesis
  - Associated with TB and lung cancer

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